

PNPInverse forward solver: changes since late April 2026

mathematical formulation, with literature provenance

2026-05-04

Abstract

This note collects the mathematical changes to the PNP–BV forward solver between the original v18/v19 log-rate landing (late April 2026) and the present production state (May 4, 2026). Each piece of mathematics is annotated with its provenance: **[LIT]** = standard result from the literature; **[APP]** = a literature result specialized to this project’s symbolic setup; **[NEW]** = a derivation that the May 2026 literature search did not place in any prior published form.

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1 Setting and notation

The active production stack tracks dynamic species $\text{O}_2, \text{H}_2\text{O}_2, \text{H}^+$ with charges $z = (0, 0, +1)$, an analytic Boltzmann counterion for ClO_4^- , log-concentration primary variables $u_i = \ln c_i$, and a log-rate Butler–Volmer (BV) boundary condition for the two-step ORR



The dimensionless potential is $\phi = \Phi/V_T$ with bulk $\phi = 0$ at $y = 1$ and the electrode at $y = 0$. The electromigration prefactor is $em \cdot z_i$; in production scaling $em \cdot z_{\text{H}^+} = +1$. A 4-species dynamic preset is also supported in which ClO_4^- ($z = -1$) is dynamic with Bikerman steric coefficient $a_{\text{ClO}_4^-} = a$.

2 Bikerman/lattice-gas steric sign correction [APP]

Origin in the literature. The free energy density of a binary lattice gas with cell size a is [1]

$$\sigma(c^+, c^-) = \frac{k_B T}{a^3} [c^+ a^3 \ln(c^+ a^3) + c^- a^3 \ln(c^- a^3) + (1 - \Phi) \ln(1 - \Phi)], \quad \Phi = a^3(c^+ + c^-). \quad (1)$$

Bazant et al. [2] write the resulting excess chemical potential of species i as

$$\mu_i^{\text{steric}} = -k_B T \ln(1 - \Phi). \quad (2)$$

Both forms are textbook [LIT].

Variational derivation. Direct partial differentiation of (1) with respect to c^+ recovers (2) via

$$\begin{aligned} \frac{\partial \sigma}{\partial c^+} &= k_B T [\ln(c^+ a^3) + 1 - \ln(1 - \Phi) - 1] \\ &= \underbrace{k_B T \ln(c^+ a^3)}_{\text{ideal}} + \underbrace{(-k_B T \ln(1 - \Phi))}_{\mu_+^{\text{steric}}}. \end{aligned} \quad (3)$$

The leading -1 from differentiating the ideal log piece cancels the $+1$ from differentiating $(1 - \Phi) \ln(1 - \Phi)$. The remaining $-\ln(1 - \Phi)$ is the excess Bikerman piece with an unambiguous *negative* sign [APP].

The bug and its fix. Until 2026-05-04, both `Forward/bv_solver/forms_logc.py:266` and the writeup `docs/PNP Equation Formulations.tex` carried $\mu^{\text{steric}} = +k_B T \ln(1 - \Phi)$, i.e. the opposite sign. That sign change makes $c_i \theta \exp(z_i \phi) = \text{const}$ the zero-flux relation, which forces counterions *larger*, not smaller, as the packing fraction tightens—no saturation, and indeed no equilibrium SS exists for the dynamic counterion above $\psi_D \approx \ln(1/(4ac_b)) \approx 4.83$ nondim ($\approx +0.124$ V vs RHE). The fix in `forms_logc.py` and `forms_logc_muh.py` is the one-character change

$$\mu^{\text{steric}} = -\ln(\theta), \quad \theta := 1 - \sum_j a_j c_j, \quad (4)$$

matched by the same flip in `scripts/verification/mms_bv_3sp_logc_boltzmann.py`. The 3sp+Boltzmann production observable shifted by +0.36% in CD/PC at $V_{\text{RHE}} = +0.66$ V; the Stern-no-Stern pinned snapshot was refreshed accordingly. Mathematics [APP]; the audit and code-spec mismatch identification are [NEW] project work.

3 Multispecies γ -corrected analytical IC [APP]

3.1 The zero-flux relation under the corrected residual

With the corrected steric chemical potential (4), the full electrochemical potential is

$$\mu_i^{\text{ch}} = \ln c_i + z_i \phi - \ln \theta. \quad (5)$$

Setting $\nabla \mu_i^{\text{ch}} = 0$ with bulk anchors $c_i(0) = c_{b,i}$, $\theta(0) = \theta_b = 1 - \sum_j a_j c_{b,j}$ gives the global constant

$$c_i \theta^{-1} e^{z_i \phi} = c_{b,i} / \theta_b. \quad (6)$$

This is the standard MPB equilibrium condition [4] [LIT].

3.2 Closed form for $c_i(\psi)$ [APP]

Let $\psi = \phi - \phi_o$ be the diffuse-layer potential measured from the outer (electroneutral) reference ϕ_o . Write $c_i(\psi) = (c_{b,i} / \theta_b) \theta(\psi) e^{-z_i \psi}$ from (6) and substitute into the closure $1 - \theta = \sum_j a_j c_j$:

$$\theta(\psi) = \frac{\theta_b}{\theta_b + S(\psi)}, \quad S(\psi) := \sum_j a_j c_{b,j} (e^{-z_j \psi} - 1). \quad (7)$$

Define the multispecies steric activity factor

$$\boxed{\gamma(\psi) = \frac{1}{1 + \sum_j a_j c_{b,j} (e^{-z_j \psi} - 1)}} \quad (8)$$

so that

$$c_i(\psi) = c_{b,i} \gamma(\psi) e^{-z_i \psi}. \quad (9)$$

This is the multispecies generalization of Borukhov's eq. (5) [1] [LIT]/[APP]; for a binary symmetric $z = \pm 1$ electrolyte, $\gamma(\psi) = [1 + 2ac_b(\cosh \psi - 1)]^{-1}$, recovering the form quoted in [4, 2].

3.3 Neutral species enter θ but drop out of $S(\psi)$ [NEW]

For a $z_j = 0$ species, $e^{-z_j\psi} - 1 = 0$, so $S(\psi)$ is *algebraically identical* to the binary symmetric case even though the lattice carries O_2 and H_2O_2 . The neutrals nevertheless do follow γ in their concentrations: $c_{\text{O}_2}(\psi) = c_{\text{O}_2,b} \gamma(\psi)$ etc. This decoupling between the algebraic form of γ and the species count n is exploited heavily by the IC: one only needs a_{H^+} and $a_{\text{ClO}_4^-}$ in the denominator, but γ multiplies every c_i . We did not find this point made explicitly in the surveyed literature; treat as [NEW] for the current 2D ORR setup.

3.4 Local-anchor variant for the analytical IC [NEW]

The IC's outer Picard loop already produces y -dependent outer profiles $H_o(y)$ etc. Outer electroneutrality $c_-(y) \approx c_b e^{\phi(y)}$ together with the chosen $\phi_o(y) = \ln(H_o(y)/c_{\text{ClO}_4^-,b})$ forces $c_{\text{ClO}_4^-,o}(y) = H_o(y)$. The IC therefore uses $H_o(y)$ as a *local* anchor for both charged species in γ :

$$\gamma_{\text{IC}}(y) = \frac{1}{1 + a_{\text{H}^+} H_o(y)(e^{-\psi(y)} - 1) + a_{\text{ClO}_4^-} H_o(y)(e^{+\psi(y)} - 1)}, \quad (10)$$

with $\psi(y)$ the matched-asymptotic Debye profile of §4. Constant-bulk anchors give a packing margin of $\approx 0.5\%$; the local-anchor variant gives $\approx 2\%$. This local construction is [NEW] (not in the surveyed Bikerman literature, which is overwhelmingly 1D with bulk anchors and no reactive transport).

3.5 The chemical potential cancels γ in ϕ [NEW]

A natural-looking but *wrong* extension is to add $\ln \gamma$ to the electrostatic potential seed. Substituting (9) into (5) gives $\mu_i^{\text{ch}} = \ln c_{b,i} + \ln \gamma + z_i \phi - \ln(\theta_b \gamma) = \ln c_{b,i} + z_i \phi - \ln \theta_b$, i.e. γ cancels between the species and the $-\ln \theta$ contribution. Adding $\ln \gamma$ to ϕ would double-count it. Correct seeding is therefore

$$u_i^{\text{IC}}(y) = \ln c_{b,i} + \ln \gamma_{\text{IC}}(y) - z_i \psi(y), \quad \phi^{\text{IC}}(y) = \phi_o(y) + \psi(y). \quad (11)$$

This subtlety triggered a GPT-review correction round; the final form above is implemented at `Forward/bv_solver/forms_logc.py:795--833` (Option 2a'). Treat as [NEW] project clarification.

4 The matched-asymptotic Debye-layer initial condition [NEW]

4.1 Outer two-rate algebraic model [NEW]

Outside the Debye layer, electroneutrality gives $c_{\text{H}^+} \approx c_{\text{ClO}_4^-,b} e^{\phi}$, hence $J_{\text{H}^+} = -D_{\text{H}^+} c_{\text{H}^+} \nabla(\ln c_{\text{H}^+} + \phi) = -2 D_{\text{H}^+} \nabla c_{\text{H}^+}$, the standard ambipolar reduction [5] [LIT]. The neutral fluxes are already pure diffusion, so for steady O_2 , H_2O_2 , and H^+ on $y \in [0, 1]$ with linear outer profiles, the reaction-plane outer values satisfy

$$O_s = O_b - R_1/D_{\text{O}_2}, \quad (12)$$

$$P_s = P_b + (R_1 - R_2)/D_{\text{H}_2\text{O}_2}, \quad (13)$$

$$H_o = H_b - (R_1 + R_2)/D_{\text{H}^+}, \quad (14)$$

where D_{H^+} is now the ambipolar value and the proton stoichiometry -2 cancels the factor of 2. The Boltzmann factor maps the outer proton concentration to the reaction plane: $H_s = H_o e^{-\psi_D}$,

with $\psi_D = \phi_e - \phi_o$, $\phi_o = \ln(H_o/c_{\text{ClO}_4^-,b})$. Define BV log-prefactors A_1, B_1, A_2 from the production rate constants and η_j values, then R_1, R_2 satisfy the linear system

$$\begin{pmatrix} 1 + A_1/D_{\text{O}_2} + B_1/D_{\text{H}_2\text{O}_2} & -B_1/D_{\text{H}_2\text{O}_2} \\ -A_2/D_{\text{H}_2\text{O}_2} & 1 + A_2/D_{\text{H}_2\text{O}_2} \end{pmatrix} \begin{pmatrix} R_1 \\ R_2 \end{pmatrix} = \begin{pmatrix} A_1 O_b - B_1 P_b \\ A_2 P_b \end{pmatrix}, \quad (15)$$

solved by Picard iteration with under-relaxation (default $\omega = 0.5$) together with refresh of $H_o, \phi_o, \psi_D, A_j, B_1$. The ambipolar reduction is [LIT]; assembling the *two*-rate ORR analytic Picard predictor is [NEW].

4.2 Inner Gouy–Chapman closed form [LIT]

The 1D Poisson–Boltzmann equation for the symmetric $\text{H}^+/\text{ClO}_4^-$ pair in nondimensional form,

$$\varepsilon \psi'' = 2H_o \sinh \psi, \quad \lambda_{D,o} = \sqrt{\varepsilon/(2H_o)}, \quad (16)$$

admits the standard Gouy–Chapman solution

$$\boxed{\psi(y) = 4 \operatorname{atanh}\left(\tanh(\psi_D/4) e^{-y/\lambda_{D,o}}\right)} \quad (17)$$

[5, 2] [LIT]. We use this profile inside the Debye layer for the IC and—together with the local-anchor γ_{IC} from (10)—obtain a matched-asymptotic IC that is internally consistent at $y = 0$ and $y = \infty$ but *not* pointwise consistent with the Bikerman-modified profile in the saturated zone (see §5).

4.3 Composite IC field [NEW]

Combining the outer linear profiles, the inner GC potential, and the γ -corrected concentration map gives the production IC fields

$$\begin{aligned} \phi^{\text{IC}}(y) &= \ln(H_o(y)/c_{\text{ClO}_4^-,b}) + \psi(y), \\ H^{\text{IC}}(y) &= H_o(y) \gamma_{\text{IC}}(y) e^{-\psi(y)}, \\ \text{ClO}_4^{-\text{IC}}(y) &= H_o(y) \gamma_{\text{IC}}(y) e^{+\psi(y)}, \\ \text{O}_2^{\text{IC}}(y) &= O_b + (O_s - O_b)(1 - y), \quad (\text{times } \gamma \text{ in 4sp branch}) \\ \text{H}_2\text{O}_2^{\text{IC}}(y) &= P_b + (P_s - P_b)(1 - y), \quad (\text{times } \gamma \text{ in 4sp branch}). \end{aligned} \quad (18)$$

Here $H_o(y)$ is the linear interpolant from $H_o(0)$ (Picard surface value) to H_b at $y = 1$. Empirical performance [NEW] (see `StudyResults/ic_refinement_study/`): on $V_{\text{RHE}} \in [+0.30, +0.60]$ V the IC reduces total Newton iterations to SS by $1.7\times - 6.3\times$ over the linear- ϕ baseline, and unblocks $V_{\text{RHE}} = +0.60$ V where direct $z = 1$ SS with linear- ϕ fails.

4.4 Why the IC silently no-ops at cathodic V [NEW]

At cathodic voltages the ORR is mass-transport-limited, so $R_1 + R_2 \approx D_{\text{H}^+} H_b$ saturates on the first Picard iterate and $H_o \rightarrow (\text{numerical floor})$. Then $\phi_o \sim -690$ and ψ_D flips sign, sending H_s to underflow. The next Picard iterate flips R_1 negative; under-relaxation $\omega = 0.5$ does not damp this bistability and Picard hits the 50-iter cap. The orchestrator falls back to linear- $\phi + z$ -ramp, which is the right behavior for mass-transport-limited regimes. Diagnosis is [NEW]; quantitative voltage breakpoint $V_{\text{RHE}} = +0.30$ V is empirical.

5 Composite asymptotic $\psi(y)$ for the Bikerman regime [APP]

5.1 Bikerman first integral

For the symmetric $z = \pm 1$ Bikerman/Kornyshev MPB, the first integral is [1, 2]

$$\left(\frac{d\psi}{dy}\right)^2 = \frac{2}{\nu \lambda_D^2} \ln[1 + \nu(\cosh \psi - 1)], \quad \nu = 2ac_b. \quad (19)$$

Standard Gouy–Chapman gives $(\psi')^2 = (2/\lambda_D^2)(\cosh \psi - 1)$. In the saturated regime $\nu(\cosh \psi - 1) \gg 1$ the Bikerman form grows like $|\psi|$ rather than $\exp|\psi|/2$, so ψ must extend deeper into the bulk to absorb a given applied voltage [LIT].

5.2 Saturation crossover and composite profile [APP]

Define the saturation potential $\psi_{\text{sat}} = \ln(2/\nu) = \ln(1/(ac_b))$, which for $a = 0.01$, $c_b = 0.2$ is $\psi_{\text{sat}} \approx 6.21$ nondim. Above ψ_{sat} the first integral linearises to $(\psi')^2 \approx (2/(\nu \lambda_D^2))(|\psi| - \psi_{\text{sat}})$; below ψ_{sat} it linearises to the standard Debye exponential. The composite asymptotic profile is

$$\psi(y) = \begin{cases} \psi_D - \alpha(\psi_D) y, & 0 \leq y \leq y_m, \\ \psi_{\text{sat}} \exp(-(y - y_m)/\lambda_D), & y \geq y_m, \end{cases} \quad (20)$$

with slope $\alpha(\psi_D) = \sqrt{(2/(\nu \lambda_D^2)) \ln[1 + \nu(\cosh \psi_D - 1)]}$ and C^0 matching at $y_m = (|\psi_D| - \psi_{\text{sat}})/\alpha$. The two limits and their matching are textbook MPB [LIT]; the particular two-zone closed form, the safety guards ($\psi_{\text{sat}}(1 - \varepsilon)$ near $|\psi_D| \approx \psi_{\text{sat}}$; $\alpha \rightarrow 0$ guarded near $\psi_D = 0$), and the integration with the production code are [APP]/[NEW].

5.3 Status (2026-05-04)

Option 2b is planned but not yet executed. The 2a' sweep (γ on every species, GC ψ , sign-fixed steric residual) already reached $V_{\text{RHE}} \leq +0.50$ V cold-converged with Stern $C_S = 0.10$, with $c_{\text{ClO}_4^-}(0) \rightarrow 1/a$ saturating cleanly. $V_{\text{RHE}} \geq +0.55$ V cold-fails despite a healthy saturated IC; this is the empirical signal that 2b's modified-Bikerman ψ is needed.

6 Steric-aware analytic counterion closure [NEW]

6.1 The reduction

The 3sp+Boltzmann production stack eliminates ClO_4^- as a dynamic NP unknown by setting $c_-(x) = c_{\text{ClO}_4^-,b} e^\phi$ in the Poisson source. This is correct *only* for $\mu_-^{\text{steric}} = 0$, i.e. when $a_{\text{ClO}_4^-} = 0$. Once we restore $\mu_-^{\text{steric}} = -\ln \theta$ from (4), the inert-counterion zero-flux condition $\nabla \mu_-^{\text{ch}} = 0$ becomes $\ln c_- - \phi - \ln \theta = \ln c_{\text{ClO}_4^-,b} - \ln \theta_b$, i.e. $c_- e^{-\phi}/\theta = c_{\text{ClO}_4^-,b}/\theta_b$. With $\theta = 1 - A_d - a_{\text{ClO}_4^-} c_-$ where $A_d = \sum_{i \in \text{dyn}} a_i c_i(x)$, this is linear in c_- :

$$c_-(x) = \frac{c_{\text{ClO}_4^-,b} e^{\phi(x)} [1 - A_d(x)]}{\theta_b + a_{\text{ClO}_4^-} c_{\text{ClO}_4^-,b} e^{\phi(x)}}. \quad (21)$$

Saturation: as $\phi \rightarrow +\infty$, $c_- \rightarrow (1 - A_d)/a_{\text{ClO}_4^-}$. At the bulk $\phi = 0$, $A_d = A_{d,b}$, the expression collapses to $c_- = c_{\text{ClO}_4^-,b}$. In the dilute limit $a_{\text{ClO}_4^-} \rightarrow 0$ with $A_d \rightarrow 0$, $c_- \rightarrow c_{\text{ClO}_4^-,b} e^\phi$ and the original ideal Boltzmann is recovered.

6.2 Provenance and what is new

The structure of (21) – a Bikerman correction applied to a single inert ion using the local dynamic-species packing fraction – did not appear in the May 2026 literature search. The two-step zero-flux algebra (§3) is [APP]/[LIT]; reading off a single-ion analytic closure that *simultaneously* (i) feeds Poisson and (ii) appears in the dynamic-species packing $\theta_{\text{tot}} = 1 - A_d - a_{\text{ClO}_4^-} c_-$ is [NEW] for this project. Both occurrences (Poisson source and θ) must use the same expression; that mathematical constraint is enforced by inlining the closure helper `Forward/bv_solver/boltzmann.py::build_steric_boltzmann_expressions` in both places.

6.3 Equivalence to dynamic 4sp Bikerman

At steady state with no ClO_4^- flux at non-bulk boundaries and (4) as the steric chemical potential, the 4sp dynamic and 3sp+steric-Boltzmann stacks solve the same PDE. The MMS test `tests/test_mms_steric_boltzmann_convergence.py` verifies an L2 convergence rate ≥ 1.95 for the manufactured-source variant, and `tests/test_steric_boltzmann_closure.py::test_steric_boltzmann_equiv_t` confirms agreement on the cathodic overlap window.

7 Hybrid proton electrochemical-potential transform μ_{H^+} [APP]

7.1 Motivation [LIT]

At Boltzmann equilibrium the electrochemical potential μ_i^{ch} is constant; in particular $u_{\text{H}^+} = \ln c_{\text{H}^+}$ and ϕ each vary by tens of log-units across the Debye layer while $\mu_{\text{H}^+}^{\text{ch}} = u_{\text{H}^+} + em z_{\text{H}^+} \phi$ is approximately flat. Slotboom’s classical change of variable [6] for semiconductor drift–diffusion uses exactly this fact; the same transformation generalizes to PNP electrolytes, where the result is sometimes called a quasi-Fermi or electrochemical-potential variable [LIT].

7.2 The hybrid implementation

We change the proton primary variable only:

$$\mu_{\text{H}^+} := u_{\text{H}^+} + em z_{\text{H}^+} \phi \iff c_{\text{H}^+} = \exp(\mu_{\text{H}^+} - em z_{\text{H}^+} \phi). \quad (22)$$

Neutrals $\text{O}_2, \text{H}_2\text{O}_2$ keep $u_i = \ln c_i$. The proton flux,

$$J_{\text{H}^+} = -D_{\text{H}^+} c_{\text{H}^+} \nabla \mu_{\text{H}^+} - D_{\text{H}^+} c_{\text{H}^+} \nabla \mu^{\text{steric}}, \quad (23)$$

is reconstructed from the new unknown plus the resolved ϕ , with no separate $em z_{\text{H}^+} \nabla \phi$ drift term. This is mathematically equivalent to the log-c form when both converge to the same SS, but gives Newton a much smoother primary variable in the Debye layer.

7.3 IC under cancellation [NEW]

Plugging the matched-asymptotic IC (11) into (22) with $em z_{\text{H}^+} = +1$,

$$\mu_{\text{H}^+}^{\text{IC}} = \ln H_o(y) - \psi(y) + \phi_o(y) + \psi(y) = \ln H_o(y) + \ln(H_o(y)/c_{\text{ClO}_4^-,b}). \quad (24)$$

The diffuse-layer ψ cancels exactly: the IC for μ_{H^+} is visibly smooth in y . The cancellation is what makes Newton easier in this variable; it is observed numerically and is [NEW] only in the sense that the explicit cancellation has not been written out in the surveyed PNP literature for this specific reactive ORR setup.

7.4 Implementation

The hybrid lives at `Forward/bv_solver/forms_logc_muh.py` (964 LOC, near-clone of `forms_logc.py`); routing is via `formulation="logc_muh"`. Five `c_H`-touch sites were substituted (NP flux, BV surface concentration, BV log-rate, Bikerman packing, Poisson source) plus the IC pair. The `u_clamp` is applied to the reconstructed $\ln c_{H^+} = \mu_{H^+} - em z_{H^+} \phi$, not to raw μ_{H^+} , because the offset by ϕ is $O(40)$ at high anodic V and a symmetric clamp on raw μ_{H^+} would impose a different physical bound. The $c_{H^+,old}$ in the time term must be reconstructed using ϕ_{prev} , not the current ϕ , or transient runs are silently inconsistent.

8 The three clipping conventions, audited [NEW]

The phrase “log-rate removed exponent clipping” in older docs is incorrect. Three distinct clamps appear in the production stack; log-rate eliminates only one.

8.1 η -clip at ± 50 (active)

$\eta_{scaled} = (V_{RHE} - E_{eq,j})/V_T$, clipped to $[-50, +50]$ *before* multiplication by αn_e [8] [NEW]. The clip is on η_{scaled} , not on $\alpha n_e \eta$; this matters because the unclipping voltage for R_2 ($E_{eq,2} = 1.78$ V, $n_e = 2$) is

$$V_{unclip}^{R_2} = E_{eq,2} - 50 V_T = +0.495 \text{ V}, \quad (25)$$

not the $+1.14$ V quoted in older handoffs (which mistakenly applied the clip to $\alpha n_e \eta$ and dropped the α factor). The corrected threshold is project-internal [NEW]. Below this voltage, R_2 ’s cathodic exponent is frozen at $\alpha_2 n_e \cdot 50$; above, the true α_2 -dependence recovers (measured $\partial CD / \partial \alpha_2$ jumps $\sim 340\times$ across threshold). The clip is *not* merely an overflow guard for the production V range; it is also a Newton-stability device on cold transients. At converged SS, $u_{cat} \approx -\alpha n |\eta|$ and the log-rate sum cancels, so the clip is effectively inactive at SS for V above its threshold.

8.2 U -clamp at ± 30 (active in bulk only)

$c_i = \exp(\text{clamp}(u_i, \pm 30))$ appears as a coefficient in bulk integrals (time term, NP flux, Poisson source). Log-rate cannot algebraically defer this $\exp$ because it is not a single scalar rate but a coefficient that multiplies test-function expressions over Ω [NEW]. For $V_{RHE} > +0.30$ V the SS $u_{H_2O_2}$ approaches -30 and the clamp begins to bind at SS; widen to $u_{clamp} = 100$ in that regime.

8.3 Surface c_{surf} clamp inside BV (removed by log-rate)

The legacy BV residual evaluates $k_0 c_{surf} \exp(-\alpha n_e \eta)$ with $c_{surf} = \exp(\text{clamp}(u, \pm 30))$. When the production V drives $c_{H_2O_2}$ below the floor, the floored c_{surf} multiplied by the saturated $\exp(\alpha n_e \cdot 50)$ produces a spurious R_2 sink. The log-rate rearrangement

$$\log r_{cath} = \ln k_0 + u_{cat} + \sum_{others} p_s (u_s - \ln c_{ref}) - \alpha n_e \eta, \quad r = \exp(\log r_{cath}) - \exp(\log r_{anod}) \quad (26)$$

defers the only $\exp$ in the rate expression to the very end and uses u_{cat} unclamped [NEW]. This eliminates the artificial sink and the floor pinning, but *does not* eliminate either of the other two clips.

8.4 CD vs PC asymmetry below threshold [NEW]

The clip-threshold sensitivity sweep (`StudyResults/clip_threshold_sensitivity/`, 2026-05-03) established that:

- CD is approximately preserved across $\text{clip} \in \{50, 100\}$ ($\sim 0.2\%$ max distortion at $V_{\text{RHE}} = -0.5$ V).
- PC is qualitatively wrong below $V_{\text{RHE}} = -0.1$ V at $\text{clip} = 50$: opposite sign and 3–4 OOM magnitude shift relative to $\text{clip} = 100$.
- Internal $c_{\text{H}_2\text{O}_2}^{\text{surface}}$ at $\text{clip} = 50$ *accumulates* above bulk; the unclipped truth depletes below bulk by 9–12 OOM.
- $\partial R_2 / \partial \alpha_2 \approx 0$ in the clipped regime because the exponent is frozen w.r.t. α_2 .

These observable-level asymmetries are the empirical motivation for restricting PC fits to $V_{\text{RHE}} \geq -0.1$ V at $\text{clip} = 50$, or re-running at $\text{clip} = 100$ (smaller cold-start basin: 10/13 $\rightarrow$ 3/13 cold-converged on the production grid, but warm-walk continuation handles the remainder).

9 Production-tree cleanup [NEW] (no math)

The pre-cleanup stack (commit 6a46d98, tagged `pre-cleanup-2026-05-02`) carried both the production 3sp+Boltzmann+log-c+log-rate path and the legacy 4sp/concentration backend that this work superseded. Phases 1–5 of the cleanup removed –9,468 LOC across 8 dead `Forward/bv_solver/` modules, 7 dead test/verification files, and 1 broken study script; the production driver `scripts/plot_iv_curve_unified` reproduced the pre-cleanup baseline *bit-exactly* on a 13-voltage sweep at $N_y = 200$. The dispatcher `Forward/bv_solver/dispatch.py` collapsed from 125 LOC to 51 LOC: `build_context`, `build_forms`, and `set_initial_conditions` are now one-line wrappers around `forms_logc {...}_logc` and the new `forms_logc_muh {...}_logc_muh`.

10 Status and pointers

Topic	Production state (2026-05-04)	Code pointer
Steric sign	corrected to $-\ln \theta$	<code>forms_logc.py:266</code>
Multispecies γ IC (4sp)	local-anchor H_o , in 4sp branch only	<code>forms_logc.py:795--833</code>
Composite ψ (Option 2b)	planned, not executed	<code>4sp_bikerman_ic_option_2b_plan.md</code>
Steric Boltzmann counterion	opt-in via <code>steric_mode="bikerman"</code>	<code>boltzmann.py::build_steric_boltzmann_expres</code>
μ_{H^+} hybrid	<code>formulation="logc_muh"</code>	<code>forms_logc_muh.py</code>
η -clip	± 50 default; widen to ± 100 for clean PC below threshold	<code>forms_logc.py::build_eta_clipped</code>
U -clamp	± 30 default; widen to ± 100 for $V > +0.30$ V	<code>forms_logc.py:194--198</code>
Tree cleanup	–9,468 LOC; bit-exact baseline	tag <code>pre-cleanup-2026-05-02</code>

Table 1: Status of the May 2026 forward-solver changes.

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- [8] Project source. `Forward/bv_solver/forms_logc.py:217--236 (_build_eta_clipped)`. See `docs/clipping_conventions.md`.